ToPP! Topology of Pulsar Profiles

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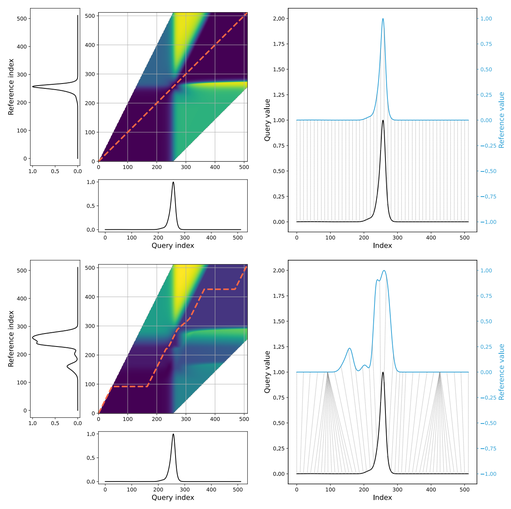
article with J. van Leeuwen & Y. Maan

Comparing profile's Stokes I morphology



$w = 0$ if identical

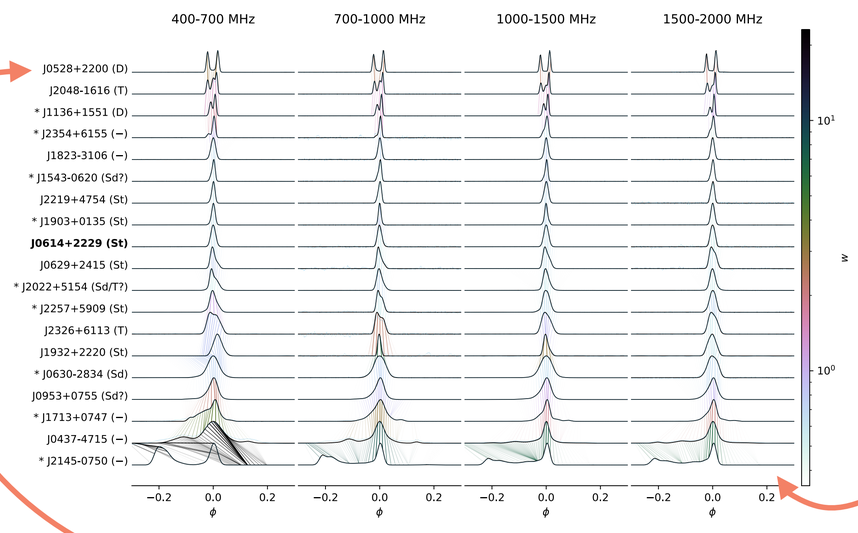
$w > 0$ if different

A **pulsar average profile** (intensity vs rotational phase) is its signature — essential tool for pulsar timing

Average profiles are (mostly) time-stable carrying info about emission regions and geometry

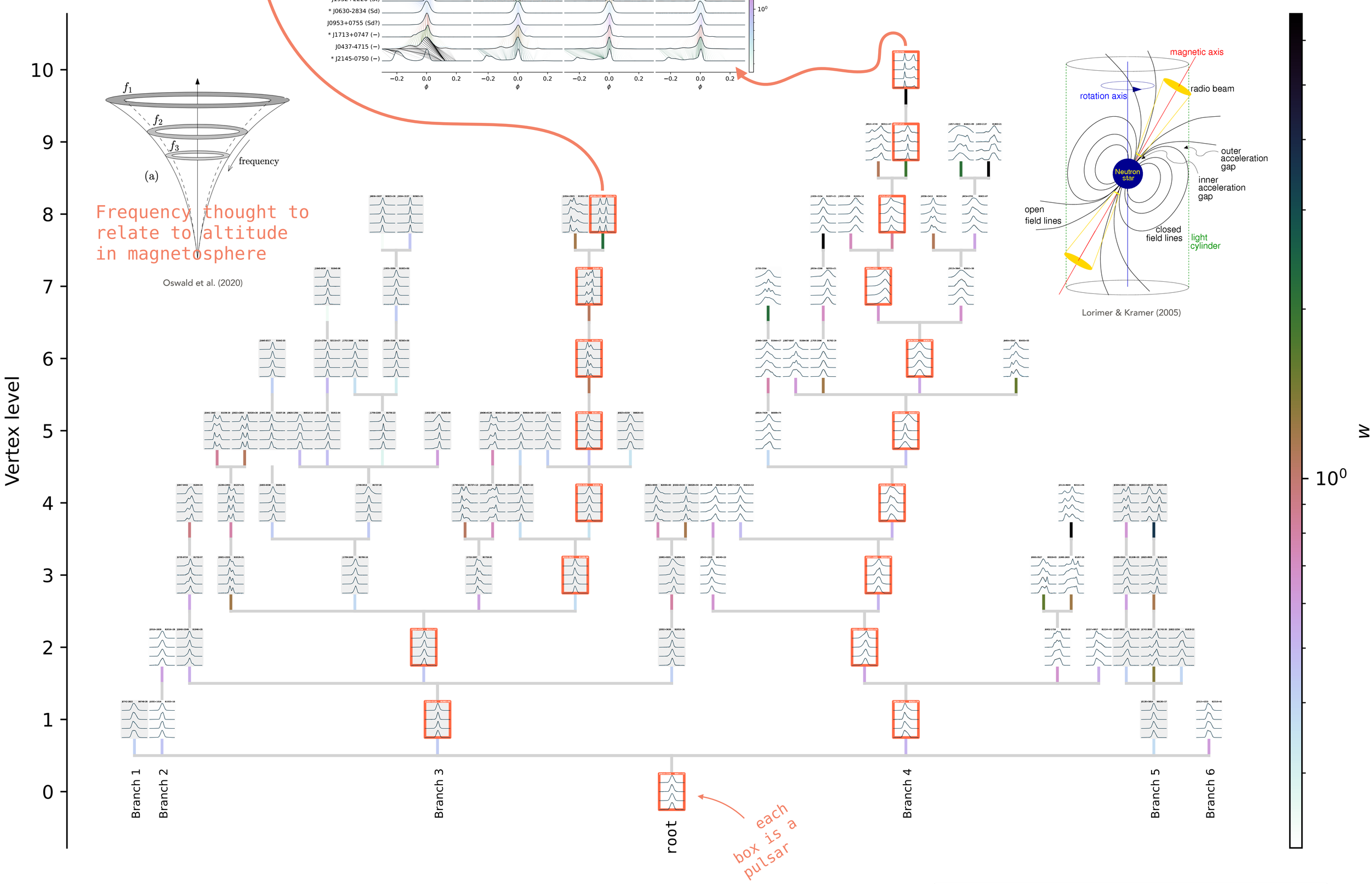
How do 90 pulsars relate?

We use graph theory to investigate the European Pulsar Network database.



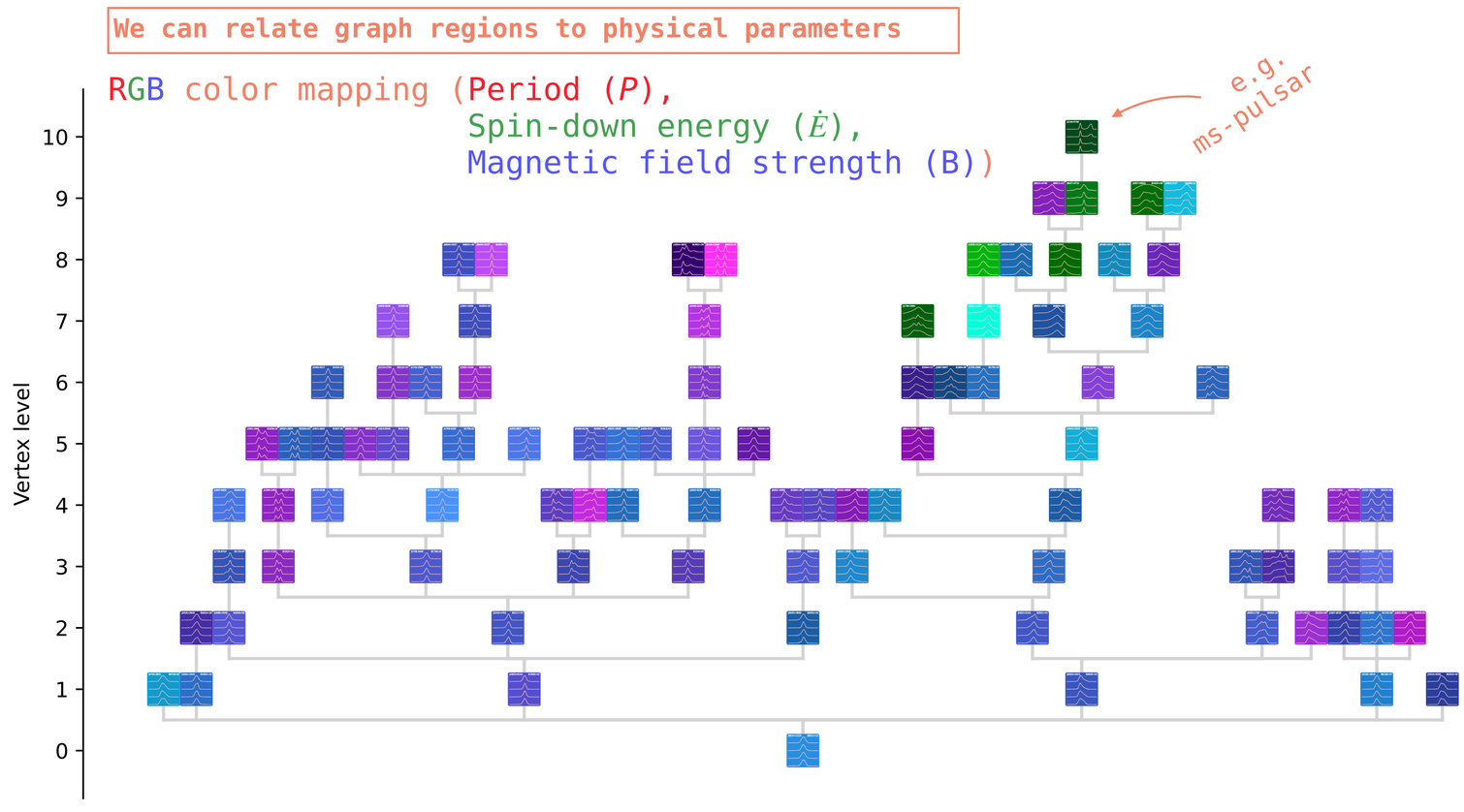
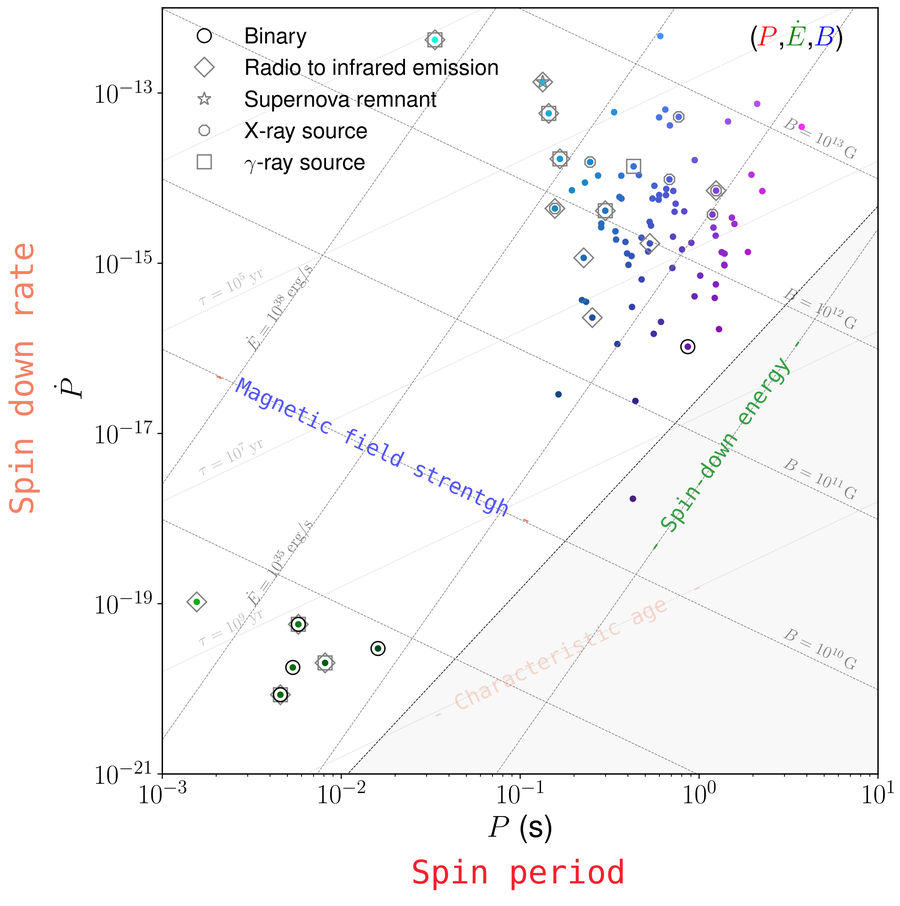
Seaquencing highlights morphology evolution

Minimum Spanning Tree on *w*
(organizes pulsars by spectro-morphology)



Frequency thought to relate to altitude in magnetosphere

each box is a pulsar

We can relate graph regions to physical parameters

RGB color mapping (Period (P), Spin-down energy ($\dot{E}$), Magnetic field strength (B))